



Diaspora: The Free, Distributed Social Network

This month, the author explores Diaspora software, which is free and can be used to set up non-profit distributed social networks. A group of independently owned nodes can interoperate to form a network.

Diaspora (wiki.diasporafoundation.org) is a Ruby on Rails application for social networking, designed to run as a federated system. Each Diaspora server is called a 'pod', and you can join one of the public pods. There are a few with the 'in' domain as well. You can select a pod using the guidelines on the Diaspora wiki or use the site <https://podupti.me/>.

In this article, we will explore how to set up a personal pod. This may be useful in an organisation or a group, as it can encourage sharing and networking between colleagues. An additional advantage is that, as it is open source, code can be added to make it more useful for any special needs of the common interest group.

Installation

The installation of Diaspora is complex but not very difficult. Instructions are available on the wiki for various Linux distributions as well as for MacOS and for Windows.

The instructions for Fedora 22 are fine for Fedora 25 as well, with some minor differences.

The following line created a little confusion:

```
curl -L https://s.diaspora.software/it | bash
```

I misread 'l' in 'It' as the letter L and not the number 1!

For testing, you can use a dummy URL and self-signed certificates. However, for production, you will need valid certificates and a valid domain. The dummy URL should be in the `/etc/hosts` file so that the browser can find it.

The configuration files are in `.yml` and, hence, indentation is as critical as in Python code.

To get the Apache reverse proxy to work, you need to make sure that `mod_ssl` is installed. Modification of the sample configuration file for Apache is fairly straightforward. You need to take care of the file names, e.g., for the proxy and SSL settings, as follows:

```
<Proxy balancer://upstream>
```

```
BalancerMember unix:///home/diaspora/diaspora/tmp/
diaspora.sock|http://
</Proxy>
# SSLEngine On
SSLCertificateFile /etc/pki/tls/certs/localhost.crt
SSLCertificateFile /etc/pki/tls/certs/ca-bundle.crt
SSLCertificateKeyFile /etc/pki/tls/certs/localhost.key
```

Once the installation is successful, you can start the browser and explore further.

Usage

Sharing concepts are very similar to how it is done on G+ (Google Plus). Relationships are asymmetric. You create 'aspects', which are identical to circles in G+. You add people you know to one or more aspects. They will be notified that you are sharing posts with them. However, it is up to them to decide whether to include you in an aspect or not. If they don't, the posts you share with them will not show up in their stream.

You can share your posts with one or more aspects, or the public. However, your post will go to a user's stream, provided this user is following you. This means that you need to be one of this user's aspects.

Re-sharing of posts is tighter. Only public posts can be re-shared. You need to keep in mind that someone may re-share your private post by using copy-paste.

In addition to seeing posts from people you follow, you may also view posts based on your interests. This is similar to the micro-blogging sites. You can specify your interests by selecting a list of hashtags. A public post that contains one of the hashtags in your list of interests will also be shown on your stream.

Networking

Diaspora pods do not synchronise, which is fine. There is no reason for a pod to keep track of all the traffic. If a user on one pod is following a user on another pod, then the relevant posts will be forwarded. However, if the receiving